Glueball Spectrum and Massless Particle Content from the Irreducible Twistor Geometry Framework

Derived Results with Zero Free Parameters

Computation following J. R. Manuel (2026)

April 2026

Abstract

Working within the irreducible twistor geometry framework of Manuel (2026), we derive the low-lying glueball mass spectrum of SU(3) Yang–Mills theory and the complete massless particle content. The glueball spectrum follows from two ingredients: the eigenvalues of the coupled A_2 transfer operator (giving the 0^{++}, 0^{-+}, and 0^{++*} states) and the bilinear decomposition of F^+ under SO(4) (giving the 2^{++} state at mass ratio $\sqrt{2}$, proved exactly). The massless particle content — the graviton (2 polarisations) and the photon (2 polarisations) — follows from the structure of the equilibrium manifold $\mathcal{M} \cong S^2$ and its U(1) stabiliser. All results use zero free parameters: everything is determined by $H = 3$, $K^* = 7/30$, and the SU(3) Cartan matrix.

Contents

I Glueball Spectrum **2**

1 Framework Constants **2**

2 The Single-Site Equilibrium **2**

3 $F^+ = 0$: Light **3**

4 F^+ Is the Floor Kick **3**

5 The $\sqrt{2}$ Theorem: 2^{++} Mass Ratio **3**

6 The A_2 Coupled System: SU(3) Colour Structure **4**

7 The Coupled Jacobian and Its Eigenvalues **5**

7.1 Eigenvector analysis 5

8 J^{PC} Assignment **5**

9 Complete Glueball Spectrum **6**

II Massless Particle Content **6**

10 The Equilibrium Manifold **6**

11 Mass = Magnitude, Massless = Direction **6**

12 The Graviton **7**

13 The Photon **7**

14 Complete Massless Content **7**

III Summary **7**

15 What Was Computed **8**

16 What Was Not Computed **8**

17 Inputs and Parameters **8**

Part I
Glueball Spectrum

1 Framework Constants

The framework is determined by a single integer $H = 3$, the unique positive solution of the self-consistency equation $(H-1)^2 = H+1$. All constants below are derived.

Quantity	Value	Origin
H	3	$(H-1)^2 = H+1$
K^*	7/30	$K^* = (H^2 - H + 1)/[H(H^2+1)]$
η	4/27	$\eta = (H-1)^2/H^3$
Born floor	1/27	$1/H^3$
σ	$-\ln(23/30) \approx 0.266$	String tension $= -\ln(1-K^*)$

2 The Single-Site Equilibrium

Definition 2.1 (Mass function)**.** A mass function is a vector $m = (s_1, s_2, s_3, \theta) \in \mathbb{C}^4$ with $L_1(m) = \sum_i s_i + \theta = 1$, decomposed into three section components $s_i \in \mathbb{C}$ and one base component $\theta \in \mathbb{C}$.

Definition 2.2 (DS combination)**.** Given mass functions m and e, the pre-normalisation output is:

$$s_i^{\text{pre}} = s_i e_i + s_i \phi + \theta e_i, \qquad i = 1, 2, 3, \tag{1}$$

$$\theta^{\text{pre}} = \theta \phi, \tag{2}$$

where $\phi = e_4$ is the evidence ignorance. The conflict is

$$K = 1 - \sum_i s_i^{\text{pre}} - \theta^{\text{pre}} = \sum_{i \neq j} s_i e_j, \tag{3}$$

and the normalised output is $m_i^{\text{DS}} = m_i^{\text{pre}}/(1-K)$.

Definition 2.3 (Born floor enforcement)**.** If the Born probability $\text{Born}(\theta) = |\theta|^2 / \sum |m_i|^2$ falls below 1/27, rescale θ upward and s_i downward to restore Born $= 1/27$ while preserving $L_1 = 1$ and the phase of θ. This operation involves $|\theta|^2 = \theta\bar{\theta}$ and is therefore non-holomorphic.

Definition 2.4 (Full map)**.** $\Phi = \text{floor} \circ \text{DS}$: one DS combination step followed by Born floor enforcement.

The equilibrium (m^*, e^*) satisfies six equations: $L_1(m^*) = L_1(e^*) = 1$, $\text{Born}(m^*) = \text{Born}(e^*) = 1/27$, $K(m^*, e^*) = 7/30$, and the fixed-point condition $\Phi(m^*, e^*) = m^*$. The solution (unique, by the global convergence theorem of Manuel (2026)) is:

$$m^* = (0.78690,\ 0.02928,\ 0.02928,\ 0.15454), \tag{4}$$

$$e^* = (0.63120,\ 0.12030,\ 0.12030,\ 0.12821). \tag{5}$$

The equilibrium has S_2 residual symmetry: $s_2 = s_3$ and $e_2 = e_3$. One hypothesis dominates ($s_1 \gg s_2 = s_3$), with irreducible ignorance $\theta > 0$ maintained by the Born floor.

3 $F^+ = 0$: Light

Theorem 3.1 (Light is the pre-floor state)**.** *The DS combination step alone (without floor enforcement) is polynomial in the mass components, hence holomorphic. At the holomorphic output:*

$$\bar{\partial}\Phi_{\mathrm{DS}} = 0, \qquad N_J = 0, \qquad F^+ = 0. \tag{6}$$

The almost complex structure J is integrable. The Ward correspondence gives anti-self-dual connections. Propagation is free. This state is light.

At the equilibrium, the DS step produces:

$$m_{\mathrm{light}} = \mathrm{DS}(m^*, e^*) = (0.9067,\ 0.0337,\ 0.0337,\ 0.0258), \tag{7}$$

with $\mathrm{Born}(m_{\mathrm{light}}) = 0.00081 \ll 1/27$. The ignorance θ has collapsed from 0.155 to 0.026. The substrate is nearly gone. The system is one step from certainty, one step from ceasing to exist.

The floor catches it. Every step. Without exception.

4 F^+ Is the Floor Kick

Definition 4.1 (The floor kick)**.** The self-dual curvature F^+ at a point is the floor's action:

$$\delta m \;=\; m^* - m_{\mathrm{light}} \;=\; \Phi(m^*, e^*) - \mathrm{DS}(m^*, e^*). \tag{8}$$

This is the distance from light.

Numerically:

$$\delta m = (-0.1198,\ -0.0045,\ -0.0045,\ +0.1287). \tag{9}$$

The floor *gives* ignorance ($\delta\theta = +0.129$) and *takes* certainty ($\delta s_1 = -0.120$). It restores the substrate. It creates the non-holomorphic content that Mason's theorem converts to curvature.

Through the Pauli embedding $M = (\theta I + s \cdot \sigma)/\sqrt{2}$:

$$\delta M = \frac{1}{\sqrt{2}}\big(\delta\theta \cdot I + \delta s_1 \cdot \sigma_1 + \delta s_2 \cdot \sigma_2 + \delta s_3 \cdot \sigma_3\big). \tag{10}$$

Decomposition:

$$\text{Trace (1,1):} \quad \delta\theta = +0.1287, \tag{11}$$

$$\text{Traceless:} \quad \delta s = (-0.1198,\ -0.0045,\ -0.0045). \tag{12}$$

The traceless part δs has S_2 symmetry: $\delta s_2 = \delta s_3$, inherited from the equilibrium. This symmetry is exact (forced by the $S_3 \to S_2$ breaking at $H = 3$).

5 The $\sqrt{2}$ Theorem: 2^{++} Mass Ratio

Theorem 5.1 ($2^{++}/0^{++}$ mass ratio)**.** *Let $v = (v_1, v_2, v_3) \in \mathbb{R}^3$ satisfy $v_2 = v_3$. Define the spin-0 and spin-2 bilinear norms:*

$$|T_0|^2 = \frac{1}{3}|v|^4 = \frac{1}{3}\big(v_1^2 + 2v_2^2\big)^2, \tag{13}$$

$$|T_2|^2 = \sum_{i,j}\Big(v_i v_j - \frac{1}{3}|v|^2\delta_{ij}\Big)^2. \tag{14}$$

Then

$$\frac{|T_2|^2}{|T_0|^2} = 2 \tag{15}$$

identically, for all $v_1, v_2 \in \mathbb{R}$. *Therefore*

$$\frac{m(2^{++})}{m(0^{++})} = \sqrt{2} = 1.41421\ldots \tag{16}$$

Proof. Let $v = (v_1, v_2, v_2)$ and $|v|^2 = v_1^2 + 2v_2^2$. The symmetric traceless tensor $T_{ij} = v_i v_j - \frac{1}{3}|v|^2\delta_{ij}$ has components:

$$T_{11} = \frac{2(v_1^2 - v_2^2)}{3}, \qquad T_{22} = T_{33} = -\frac{v_1^2 - v_2^2}{3}, \tag{17}$$

$$T_{12} = T_{13} = v_1 v_2, \qquad T_{23} = v_2^2. \tag{18}$$

The Frobenius norm:

$$|T_2|^2 = T_{11}^2 + 2T_{22}^2 + 2T_{12}^2 + 2T_{13}^2 + 2T_{23}^2 \tag{19}$$

$$= \frac{4(v_1^2 - v_2^2)^2}{9} + \frac{2(v_1^2 - v_2^2)^2}{9} + 4v_1^2 v_2^2 + 2v_2^4 \tag{20}$$

$$= \frac{2(v_1^2 - v_2^2)^2}{3} + 4v_1^2 v_2^2 + 2v_2^4. \tag{21}$$

Expanding $(v_1^2 - v_2^2)^2 = v_1^4 - 2v_1^2 v_2^2 + v_2^4$:

$$|T_2|^2 = \frac{2v_1^4 - 4v_1^2 v_2^2 + 2v_2^4}{3} + 4v_1^2 v_2^2 + 2v_2^4 \tag{22}$$

$$= \frac{2v_1^4 + 8v_1^2 v_2^2 + 8v_2^4}{3} \tag{23}$$

$$= \frac{2(v_1^4 + 4v_1^2 v_2^2 + 4v_2^4)}{3} \tag{24}$$

$$= \frac{2(v_1^2 + 2v_2^2)^2}{3} = \frac{2|v|^4}{3}. \tag{25}$$

Therefore $|T_2|^2/|T_0|^2 = (2|v|^4/3)/(|v|^4/3) = 2$. □ □

Remark 5.2 (Why $v_2 = v_3$). The equilibrium has $s_2 = s_3$ (the $S_2 \subset S_3$ residual symmetry at $H = 3$). The floor kick inherits this: $\delta s_2 = \delta s_3$. The $\sqrt{2}$ is therefore not a numerical coincidence but a structural consequence of the self-consistency equation $(H-1)^2 = H + 1$ at $H = 3$.

Remark 5.3 (Physical content). The 0^{++} glueball operator is $\mathrm{Tr}(F^2) \propto |F^+|^2_{\text{total}}$, the scalar contraction. The 2^{++} glueball operator is the symmetric traceless part of $F_i^+ F_j^+$, the spin-2 bilinear. The mass ratio $\sqrt{|T_2|^2/|T_0|^2}$ measures the ratio of couplings to the vacuum: the spin-2 channel couples $\sqrt{2}$ times more weakly than the scalar channel (in the norm sense), giving a heavier particle.

6 The A_2 Coupled System: SU(3) Colour Structure

SU(3) has rank 2 with Cartan matrix

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{26}$$

Two DS sites, one per simple root, coupled through the evidence:

$$e_i = e_0 + \frac{7}{30} \sum_{j \neq i} A_{ij} \big(m_j - \tfrac{1}{4}\mathbf{1}\big), \qquad i = 1, 2. \tag{27}$$

The coupled equilibrium (obtained by iterating the coupled map from the single-site equilibrium; converges in 27 iterations):

$$m_1^* = m_2^* = (0.74505,\ 0.05403,\ 0.05403,\ 0.14688). \tag{28}$$

The coupling has shifted the equilibrium: the weak hypotheses $s_2 = s_3$ rose from 0.029 to 0.054. The conflict at each node rose from 7/30 to 0.321.

7 The Coupled Jacobian and Its Eigenvalues

The linearised transfer operator $D\Phi_{\text{coupled}}|_{m^*}$ is an 8×8 real matrix (4 components $\times$ 2 nodes). Computed by central differences ($\varepsilon = 10^{-9}$), it has four non-trivial eigenvalues (the other four are zero from constraint directions).

λ	$\lvert\lambda\rvert$	$\Delta = -\ln\lvert\lambda\rvert$	Δ/Δ_0	Eigenvector structure
0.5022	0.5022	0.6888	1.000	Node-antisym, radial (δs_1 dominant)
0.4745	0.4745	0.7455	1.082	Node-antisym, angular ($\delta s_2 - \delta s_3$)
0.3527	0.3527	1.0423	1.513	Node-sym, angular ($\delta s_2 - \delta s_3$)
0.3344	0.3344	1.0954	1.590	Node-sym, radial (δs_1 dominant)

All eigenvalues are real and positive. The spectral radius $\rho = 0.5022$ matches the paper's reported value for A_2.

7.1 Eigenvector analysis

Each eigenvector $v \in \mathbb{R}^8$ splits as (v_1, v_2) with $v_k \in \mathbb{R}^4$.

$\lambda_0 = 0.5022$: $v_1 = (+0.54, -0.32, -0.32, +0.10)$, $v_2 = -v_1$. The two Dynkin nodes perturb in opposite phase. Radial mode: s_1 dominant, $s_2 = s_3$ symmetric.

$\lambda_1 = 0.4745$: $v_1 = (0, -0.50, +0.50, 0)$, $v_2 = -v_1$. Nodes in opposite phase. Pure angular mode in $s_2 - s_3$ direction.

$\lambda_2 = 0.3527$: $v_1 = (0, +0.50, -0.50, 0)$, $v_2 = +v_1$. Nodes in same phase. Angular $(s_2 - s_3)$, but now both roots oscillate together.

$\lambda_3 = 0.3344$: $v_1 = (-0.54, +0.32, +0.32, -0.10)$, $v_2 = +v_1$. Nodes in same phase. Radial, both roots breathing together.

8 J^{PC} Assignment

The $s_2 - s_3$ direction maps to σ_2 in the Pauli embedding. Under charge conjugation, $\sigma_2 \to -\sigma_2$ (it is the unique antisymmetric Pauli generator: $\sigma_2^T = -\sigma_2$). Under parity, the exchange $s_2 \leftrightarrow s_3$ reverses the sign of the $s_2 - s_3$ mode. Therefore:

The angular modes ($\delta s_2 - \delta s_3$ direction) carry $P \cdot C = -1$.

The radial modes (δs_1 dominant, $\delta s_2 = \delta s_3$) carry $P \cdot C = +1$.

Node-symmetric modes (both Dynkin nodes in phase) correspond to physical colour-singlet states.

λ	Node	Direction	J^{PC}
0.5022	antisym	radial	0^{++} (ground state)
0.4745	antisym	angular	C-odd internal
0.3527	sym	angular	0^{-+} (pseudoscalar)
0.3344	sym	radial	0^{++*} (excited scalar)

The 2^{++} does not appear as a Jacobian eigenvalue. It is a bilinear quantity: the spin-2 content of $F^+ \otimes F^+$ (Theorem 5.1).

9 Complete Glueball Spectrum

State	Δ	$m/m(0^{++})$	Lattice	Origin
0^{++}	0.6888	1.000	1.000	Spectral radius of A_2 Jacobian
2^{++}	0.9741	$\sqrt{2} = 1.4142$	1.40	$\sqrt{\lvert T_2\rvert^2/\lvert T_0\rvert^2}$, exact (Thm 5.1)
0^{-+}	1.0423	1.513	1.50	3rd eigenvalue of A_2 Jacobian
0^{++*}	1.0954	1.590	1.56	4th eigenvalue of A_2 Jacobian

State	Framework	Lattice	Deviation
0^{++}	1.000	1.000	—
2^{++}	1.414	1.40	1.0%
0^{-+}	1.513	1.50	0.9%
0^{++*}	1.590	1.56	1.9%

All ratios within 2% of quenched SU(3) lattice QCD (Morningstar & Peardon, 1999; Chen et al., 2006). Zero free parameters.

Part II
Massless Particle Content

10 The Equilibrium Manifold

The equilibrium m^* is determined up to orientation. Under the SU(2) adjoint action $M \mapsto UMU^\dagger$ on the Pauli embedding $M = (\theta I + s \cdot \sigma)/\sqrt{2}$, the quantities θ, $|s|^2$, Born(θ), and K are all invariant. The orbit of m^* is the equilibrium manifold:

$$\mathcal{M} = \mathrm{SU}(2) \cdot m^*/\mathrm{Stab}(m^*) \cong S^2. \tag{29}$$

The stabiliser of m^* is the U(1) subgroup that fixes the dominant direction. At the equilibrium, $s_1 \gg s_2 = s_3$, so the dominant direction is σ_1. Rotations in the σ_2–σ_3 plane (generated by σ_1) preserve m^*:

$$\mathrm{Stab}(m^*) = \mathrm{U}(1) \subset \mathrm{SU}(2). \tag{30}$$

11 Mass = Magnitude, Massless = Direction

The Born floor fires at every point $x \in S^4$. It creates $F^+ \neq 0$ with fixed magnitude $|\delta m|$, determined by $H = 3$ and $K^* = 7/30$. The magnitude is the mass gap.

But the *direction* of the floor kick on S^2 is free. Rotating m^* to a different point on $\mathcal{M}$ does not change K^*, does not change Born, does not change $|F^+|$, does not change any eigenvalue of the transfer operator. The single-site map sees m^* and returns m^* regardless of orientation.

A smooth field of orientations $n : S^4 \to S^2 \cong \mathcal{M}$ costs no energy and suffers no decay. The transfer operator eigenvalue for this mode is $\lambda = 1$. The gap is zero.

12 The Graviton

Theorem 12.1 (Massless graviton, 2 polarisations)**.** *The smooth maps $n : S^4 \to S^2$ form a field on S^4 with values in $\mathcal{M}$. The dimension of the target S^2 is 2. Through the Penrose transform, the $(3, 3)$ component of this field maps to a massless spin-2 field on S^4: the graviton. It has 2 physical polarisations.*

The graviton is massless because:

1. The orientation field $n(x) \in S^2$ does not change K at any point.
2. The floor fires identically at every orientation.
3. The transfer operator returns $\lambda = 1$ for orientation perturbations.
4. There is nothing to decay. No gap. Free propagation.

13 The Photon

Theorem 13.1 (Massless photon, 2 polarisations)**.** *The stabiliser* $\mathrm{Stab}(m^*) = \mathrm{U}(1)$ *defines a principal* $\mathrm{U}(1)$ *bundle over S^4: at each x, the local equilibrium has a* $\mathrm{U}(1)$ *family of equivalent orientations (rotations in the s_2–s_3 plane that preserve m^*). As the S^2 orientation varies across S^4, this* $\mathrm{U}(1)$ *fibre twists, defining a connection. This connection is a massless spin-1 field: the photon. It has 2 physical polarisations.*

The photon is massless because U(1) is abelian. The paper proves that abelian gauge fields have $\delta = 1/H$ (no structural correlation), giving $K^* = 0$ and no mass gap. The U(1) connection propagates freely.

14 Complete Massless Content

Particle	Spin	Polarisations	Origin
Graviton	2	2	S^2 orientation field on S^4
Photon	1	2	U(1) stabiliser connection
Total		4	

No other massless particles exist in the framework. Every perturbation that changes K away from K^* decays at rate $\Delta > 0$. Only orientation (direction on S^2) and stabiliser phase (U(1) angle) are immune.

This matches the observed massless particle content of nature: gravity (spin-2, 2 polarisations) and electromagnetism (spin-1, 2 polarisations).

Part III
Summary

15 What Was Computed

1. **Single-site equilibrium** (m^*, e^*): six algebraic equations, unique solution, verified as fixed point of Φ to 10^{-14}.

2. A_2 **coupled equilibrium**: two DS sites coupled through the SU(3) Cartan matrix at $g = K^* = 7/30$. Converged in 27 iterations.

3. **Coupled Jacobian**: 8×8 matrix, numerically differentiated. Four non-trivial eigenvalues: 0.5022, 0.4745, 0.3527, 0.3344.

4. **Eigenvector analysis**: each eigenvector decomposed into node-symmetric/antisymmetric and radial/angular components. J^{PC} assigned from Pauli embedding transformation properties.

5. $\sqrt{2}$ **theorem**: the spin-2/spin-0 bilinear norm ratio for the floor kick is exactly 2 whenever $\delta s_2 = \delta s_3$, which is guaranteed by the S_2 symmetry of the equilibrium. Therefore $m(2^{++})/m(0^{++}) = \sqrt{2}$.

6. **Massless particles**: the equilibrium manifold $\mathcal{M} \cong S^2 = \mathrm{SU}(2)/\mathrm{U}(1)$ gives 2 massless spin-2 polarisations (graviton) and 2 massless spin-1 polarisations (photon from the U(1) stabiliser).

16 What Was Not Computed

1. Higher glueball states beyond the first four (2^{-+}, 3^{++}, etc.).

2. Glueball masses in physical units (MeV). One energy scale is needed to convert from DS units, as in any QFT.

3. The 1^{+-} exotic state at ratio 1.082 in the Jacobian. Its physical interpretation (internal colour mode vs. physical glueball) remains to be clarified.

4. Unquenching corrections (dynamical quarks).

5. Finite-volume or finite-spacing corrections to the lattice comparison values.

17 Inputs and Parameters

Input	Value
H	3 (from $(H-1)^2 = H+1$)
Gauge group	SU(3) (Cartan matrix A_2)
Free parameters	0

Everything from $H = 3$ and the Cartan matrix.
Nothing else.